Natural Language Processing (A Classical Approach)

Certificate Course on Artificial Intelligence & Deep Learning
by IIT Roorkee

Natural Language Processing

Natural-language processing (NLP)

- An area of computer science and artificial intelligence for
- Interactions between computers and human languages
- Program computers to process natural language data

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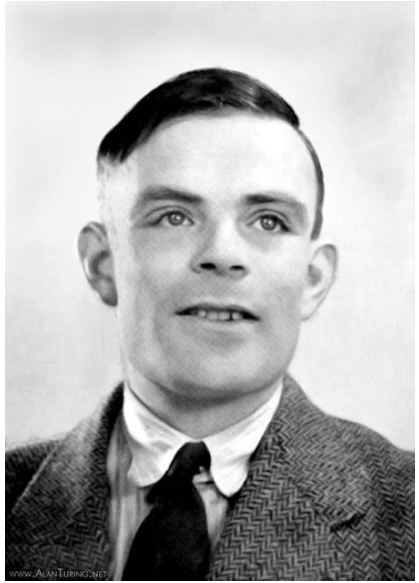
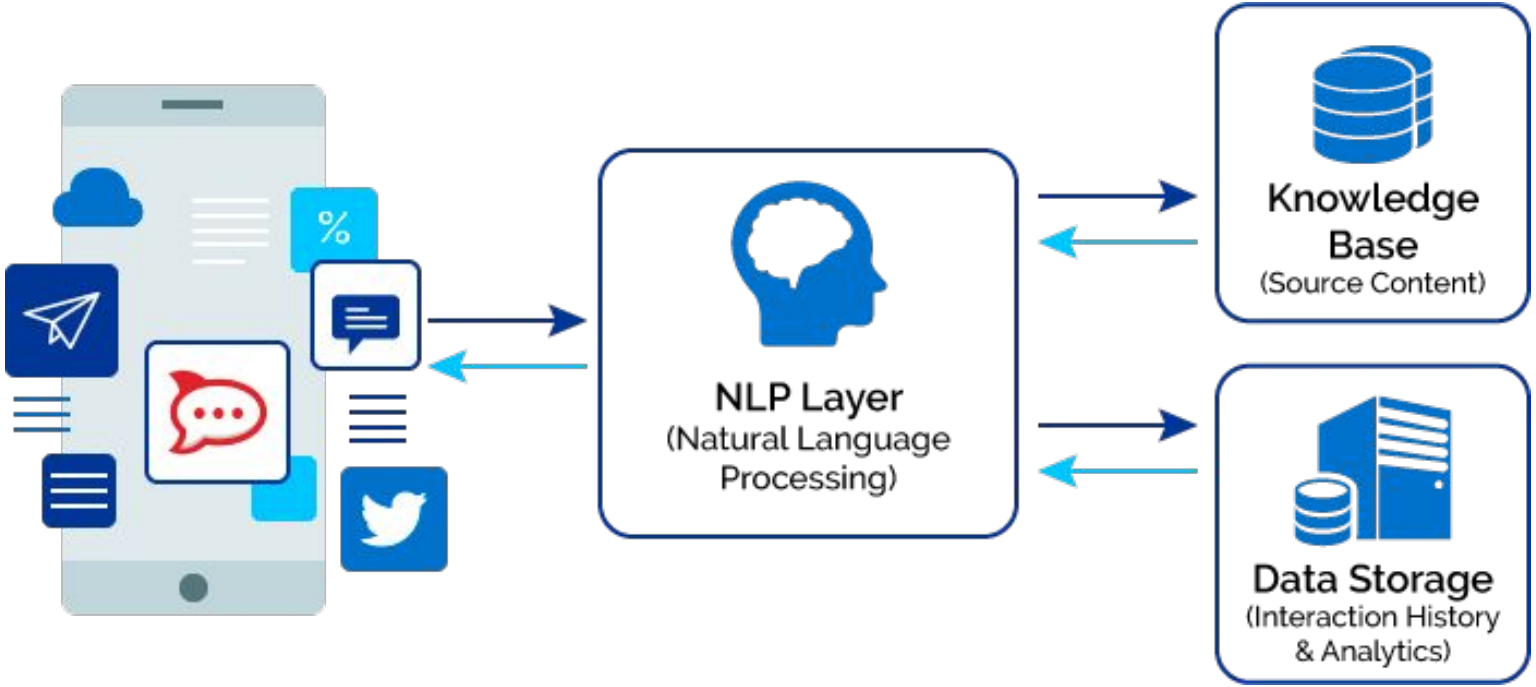
Natural Language Processing

In 1950, Alan Turing published an article titled "Computing Machinery and Intelligence" which proposed what is now called the Turing test as a criterion of intelligence.



Natural Language Processing

Basic Structure of a NLP application (chatbot considered below)



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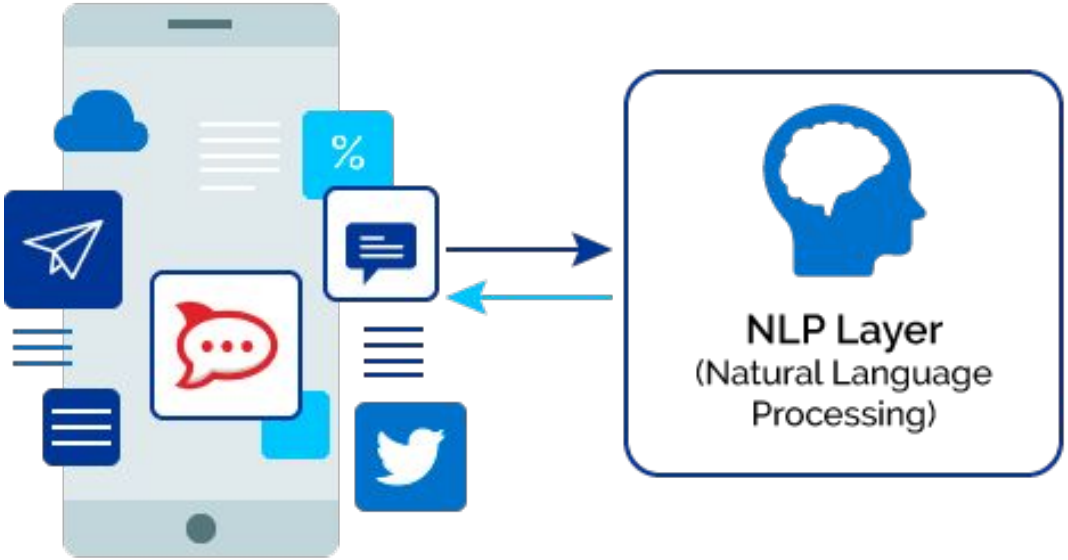
Natural Language Processing

Knowledge Base – A database used to equip chatbots with the information needed to respond to queries of customers request.

Data Store – Contains interaction history of chatbot with users.

Natural Language Processing



NLP Layer – Translates users' queries (free form) into information that can be used for appropriate responses.

Application Layer – The application interface that is used to interact with the user.

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Natural Language Processing – Applications

Speech Recognition

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Natural Language Processing – Applications

Text Classification

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Natural Language Processing – Applications

Caption Generation using VQA (Visual Question Answering)

1. Choose an Image and a Question

Kitchen

Image:

Question:

What is in the bowls on the island?

2. Run a model

Predicted Answers

Score	Answer
45.7%	bread
36.7%	fruit
5.1%	food
3.2%	cereal
1.9%	bananas

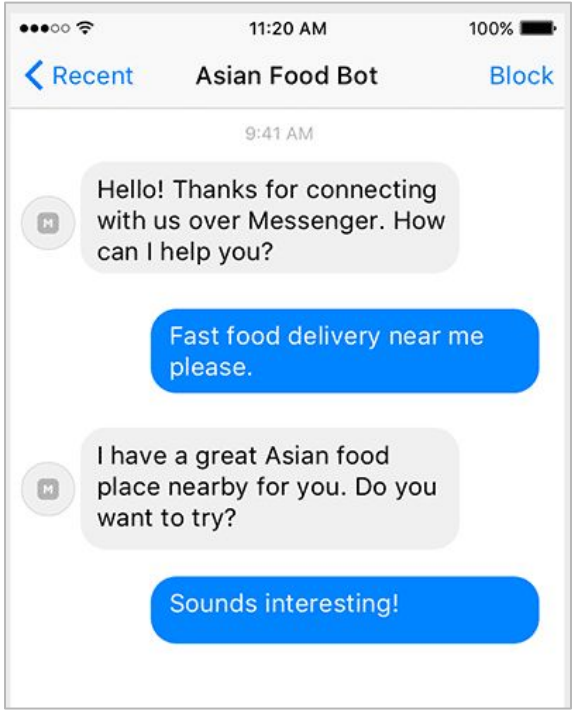
Source: https://vision-explorer.allenai.org/visual_question

Natural Language Processing – Applications

Machine Translation

Natural Language Processing – Applications

Question Answering



Natural Language Processing – Tools

Here are some of the most popular Natural Language Processing Tools.

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Natural Language Processing – Tools

- Stanford's Core NLP Suite – a set of stable and well-tested natural language processing tools, widely used by various groups in academia, industry, and government.

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Natural Language Processing – Tools

- Natural Language Toolkit (NLTK) – A suite of libraries and programs for symbolic and statistical natural language processing for English.

NLTK 3.5 documentation

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Natural Language Toolkit

NLTK is a leading platform for building Python programs to work with human language data. It provides easy-to-use interfaces to over 50 corpora and lexical resources such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries, and an active discussion forum.

Thanks to a hands-on guide introducing programming fundamentals alongside topics in computational linguistics, plus comprehensive API documentation, NLTK is suitable for linguists, engineers, students, educators, researchers, and industry users alike. NLTK is available for Windows, Mac OS X, and Linux. Best of all, NLTK is a free, open source, community-driven project.

NLTK has been called "a wonderful tool for teaching, and working in, computational linguistics using Python," and "an amazing library to play with natural language."

Natural Language Processing with Python provides a practical introduction to programming for language processing. Written by the creators of NLTK, it guides the reader through the fundamentals of writing Python programs, working with corpora, categorizing text, analyzing linguistic structure, and more. The online version of the book has been been updated for

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Natural Language Processing – Tools

- Text Blob – A Python library for processing textual data and NLP tasks like part-of-speech tagging, noun phrase extraction, sentiment analysis, and more.

TextBlob

Natural Language Processing – Tools

- SpaCy – An open-source software library for advanced natural language processing.

  - Non-destructive tokenization
  - Named entity recognition
  - "Alpha tokenization" support for over 50 languages
  - Statistical models for 11 languages
  - Pre-trained word vectors
  - Part-of-speech tagging
  - Labelled dependency parsing
  - Syntax-driven sentence segmentation
  - Text classification
  - Built-in visualizers for syntax and named entities
  - Deep learning integration

Natural Language Processing – Tools

- Pytorch-NLP – Library for NLP in Python with pre-trained embeddings, samplers, dataset loaders, metrics, neural network modules and text encoders.

Natural Language Processing – Tools

- OpenNLP by Apache – A machine learning based toolkit for the processing of natural language text.

Create a Quiz using TextBlob

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Create a Quiz using TextBlob

Let us use the TextBlob library of Python to Build a program that makes a quiz out of a provided text.

It is basically a usage of **NER** – Named-Entity Recognition.

TextBlob

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Create a Quiz using TextBlob

Let us understand a few things about the TextBlob API:

- **text.sentences** – gives the sentences in a text
- **sentences.tags** – gives the tags for each of the word in sentence (//sentences.tags returns a list of tuples with the word being the first element of the tuple and the tag being the second.)

Create a Quiz using TextBlob

Now to generate our quiz we will:

- Extract each sentence
- Replace some the nouns and proper nouns with a blank
- And remove only after the fourth word in the sentence

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Find Related Posts using Scikit-Learn

Find Related Posts using Scikit-Learn

To find related posts from a bunch of posts, we first need to learn how to turn text into something on which we can calculate similarity.

How to do it ??

By using the Bag of Words Approach

Find Related Posts using Scikit-Learn

Bag of Words Approach – It ignores order of words, and uses word counts as their basis.

In this model, a text (sentence or document) is represented as bag (multiset) of its words, disregarding grammar and word order.

Find Related Posts using Scikit-Learn

Vectorization – For each word in the post, its occurrence is counted and noted in a vector. This step is called vectorization.

The vector is typically huge as it contains as many elements as words occur in the whole dataset. //Call it vocabulary size

Vector Similarity

Vectorization example – For the two statements "How to format my hard disk" and " Hard disk format problems " the vectors are shown below in a

Term Document Matrix

Word	Occurence in post 1	Occurence in post 2
disk	1	1
format	1	1
how	1	0
hard	1	1
my	1	0
problems	0	1
to	1	0

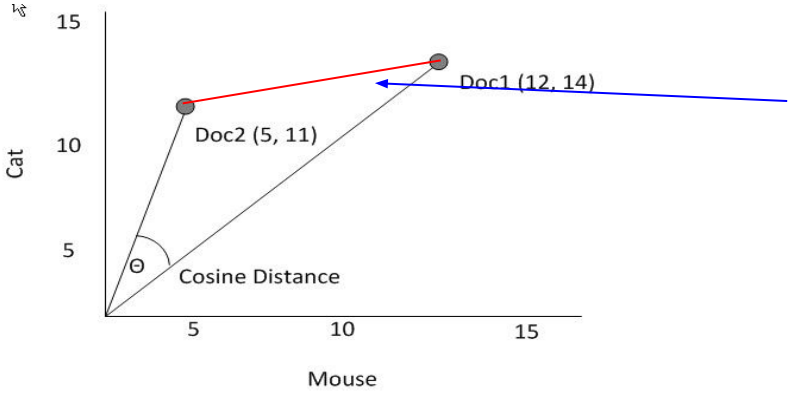
$$d(\mathbf{p},\mathbf{q}) = d(\mathbf{q},\mathbf{p}) = \sqrt{(q_1-p_1)^2+(q_2-p_2)^2+\cdots+(q_n-p_n)^2}$$

$$= \sqrt{\sum_{i=1}^{n}(q_i-p_i)^2}.$$

Cosine Similarity:

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Find Related Posts using Scikit-Learn



Euclidean Distance (Pythagorean distance):

$$d(\mathbf{p},\mathbf{q}) = d(\mathbf{q},\mathbf{p}) = \sqrt{(q_1-p_1)^2+(q_2-p_2)^2+\cdots+(q_n-p_n)^2}$$

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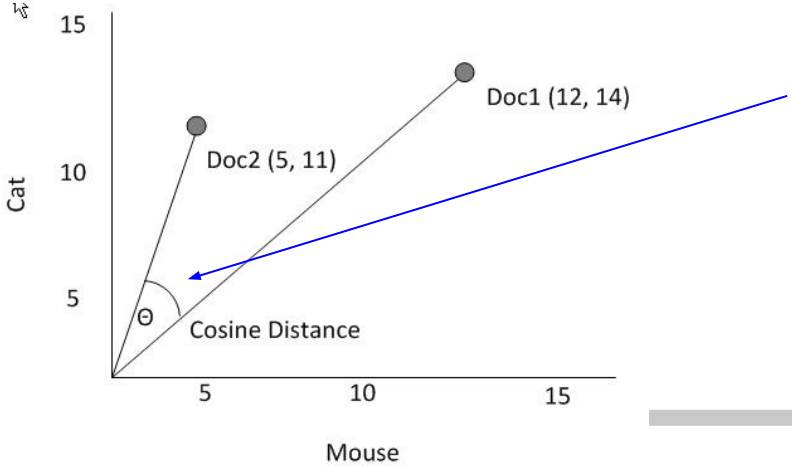

Cosine Similarity:

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Find Related Posts using Scikit-Learn

Now let's see how to implement vectorization using Scikit-Learn

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Analyse collection of Text Documents using Scikit Learn

Analyse collection of Text Documents using Scikit Learn

Here is a quick checklist of what we would do to analyse these collection of text documents:

- Load file contents and categories
- Extract feature vectors suitable for machine learning
- Train linear model to perform categorization
- Use grid search to find good configuration of feature extraction components and the classifier

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Analyse collection of Text Documents using Scikit Learn

- Occurrence count is a good start but there is an issue
  - Longer documents will have higher average count values than shorter documents

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Analyse collection of Text Documents using Scikit Learn

To avoid these potential discrepancies we:

- Divide the number of occurrences of each word in a document by the total number of words in the document:
  - These new features are called **tf for Term Frequencies**.
    - Words such as ‘is’, ‘am’, ‘that’, etc will still get higher feature value, while not carrying much information for document classification.

Analyse collection of Text Documents using Scikit Learn

How can we improve tf ?

Analyse collection of Text Documents using Scikit Learn

- Downscale feature value for words that occur in many documents in the corpus and are therefore less informative than those that occur only in a smaller portion of the corpus.
    - This downscaling is called **tf–idf for "Term Frequency times Inverse Document Frequency"**.

Analyse collection of Text Documents using Scikit Learn

- Tf-idf weight is often used in
  - Information retrieval and
  - Text mining
- This weight is a used to evaluate
  - How important a word is to a
  - Document in a collection or corpus

Analyse collection of Text Documents using Scikit Learn

Term Frequency

- Measures how frequently a term occurs in a document
- It is possible that a term would appear
  - Much more times in long documents than shorter ones
  - This is why we normalize TF

TF(t) = (Number of times term t appears in a document) / (Total number of terms in the document)

Analyse collection of Text Documents using Scikit Learn

Inverse Document Frequency

- Measures how important a term is
- In TF, all terms are considered equally important within the document
- How ever some words and stop words appears lot of time
  - But in reality have least importance for classification
- In IDF we weight down frequent terms
  - And scale up rare terms

IDF(t) = log_e(Total number of documents / Number of documents with term t in it)

Analyse collection of Text Documents using Scikit Learn

Example

- Consider a document containing 100 words and the word cat appears 3 times
- The term frequency(tf) for cat is
  - (3 / 100) = 0.03

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Analyse collection of Text Documents using Scikit Learn

Example

- Now, assume we have 10 million documents and
  - The word cat appears in 1,000 of these
- The inverse document frequency(idf) is
  - log(10,000,000 / 1,000) = 4
- Tf-idf weight is the product of tf and idf
  - 0.03 * 4 = 0.12

Analyse collection of Text Documents using Scikit Learn

Now let us apply tf-idf to our example

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Natural Language Processing – Stanford NLP

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Overview of Stanford Core NLP

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Natural Language Processing – Stanford NLP

Stanford CoreNLP provides a set of human language technology tools.

It provides:

- Base forms of words,
- Parts of speech,
- Whether they are proper nouns,

Morphological Analysis

- Mark up the structure of sentences in terms of phrases and syntactic dependencies (Parsing of the sentence or Syntactic Analysis),
- Indicate which noun phrases refer to the same entities (coreference resolution), indicate sentiment (sentiment analysis).

Natural Language Processing – Stanford NLP

Choose Stanford CoreNLP if you need:

- Integrated NLP toolkit
- Fast, robust annotator for arbitrary texts
- Highest quality of text analytics
- Support for major languages
- Available APIs for most programming languages
- Ability to run as a simple web service

Programming languages and operating systems

Stanford CoreNLP can be used via:

- Command-line
- Object-oriented simple API,
- Third party APIs in other languages
- A web service

It works on Linux, macOS, and Windows.

Other NLP Tools and Techniques to explore

Other NLP Tools and Techniques to explore

Word2Vec

The word2vec algorithm uses a neural network model to learn word associations from a large corpus of text.

Here is a Keras implementation of this technique by **Francois Chollet**.

Other NLP Tools and Techniques to explore

DialogFlow

This is an advanced development suite for creating conversational AI applications, including chatbots, voicebots, and IVR bots from Google.

Create your own projects here.

Other NLP Tools and Techniques to explore

NLP using RNN

A common and modern approach for natural language tasks is to use recurrent neural networks, we will explore them next!

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Questions?

https://discuss.cloudxlab.com

reachus@cloudxlab.com